

HORN, SwedenWMO: **025570**Lat: **57.8861N**Lon: **15.8623E**Elev: **91**StdP: **100.24**Time Zone: **1.00 (EUC)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.9	-16.2	-22.7	0.5	-20.8	-17.8	0.8	-16.0	10.6	5.2	9.3	5.1	0.1	10	0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	27.1	18.6	25.3	18.1	23.5	17.1	20.2	24.6	19.1	23.4	18.0	22.1	3.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.7	13.7	21.9	17.6	12.7	20.9	16.4	11.8	19.8	58.4	24.6	54.5	23.5	51.2	22.1	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.2	6.3	-23.5	29.4	5.7	1.6	-27.6	30.6	-31.0	31.5	-34.1	32.5	-38.3	33.7		
(5)			-24.0	21.8	5.5	1.8	-28.0	23.1	-31.2	24.2	-34.2	25.2	-38.2	26.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.3	-3.0	-2.4	0.6	5.0	9.9	13.8	16.2	15.2	11.1	6.4	2.7	-0.7
	DBStd	7.76	5.68	5.29	4.86	3.44	3.27	2.84	2.60	2.62	3.07	3.91	3.91	5.51
	HDD10.0	1933	402	348	292	152	42	3	0	1	23	121	219	331
	HDD18.3	4414	660	581	550	399	262	139	78	102	216	369	468	589
	CDD10.0	579	0	0	0	3	39	117	191	162	57	10	1	0
	CDD18.3	18	0	0	0	0	0	3	11	5	0	0	0	0
	CDH23.3	423	0	0	0	0	16	93	225	84	5	0	0	0
	CDH26.7	68	0	0	0	0	1	13	43	12	0	0	0	0
Wind	WSAvg	2.7	2.8	2.8	2.8	2.6	2.6	2.6	2.4	2.4	2.6	2.7	2.8	2.8
Precipitation	PrecAvg	617	35	33	27	34	52	69	88	74	45	55	55	50
	PrecMax	745	63	60	43	81	90	131	164	149	117	119	104	77
	PrecMin	514	5	7	3	2	16	23	27	10	8	14	14	19
	PrecStd	66	19	13	11	22	20	27	34	35	27	28	26	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.4	9.6	15.2	20.7	25.2	28.2	29.8	27.9	23.7	18.1	12.2	11.1
		MCWB	7.0	6.9	8.7	12.5	15.6	18.1	19.3	19.2	18.2	13.8	10.7	9.4
	2%	DB	6.4	7.3	12.4	18.0	22.7	26.0	27.8	25.5	21.3	15.4	10.7	9.1
		MCWB	5.1	5.3	7.5	10.9	14.4	17.4	19.0	19.6	16.4	13.1	9.6	7.6
	5%	DB	5.2	5.8	9.6	15.4	20.6	23.9	25.9	23.5	19.2	13.9	9.5	7.3
		MCWB	3.9	4.1	6.1	9.5	13.6	16.5	18.5	18.1	15.0	11.9	8.5	6.1
	10%	DB	3.9	4.4	7.5	13.0	18.3	21.8	24.0	21.8	17.3	12.6	8.2	5.8
		MCWB	2.9	3.0	5.0	8.1	12.5	15.1	17.8	16.9	14.0	10.9	7.4	4.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.3	7.5	9.8	13.6	17.3	20.2	21.3	22.8	19.0	14.7	11.2	9.6
		MCDB	8.3	8.9	13.6	20.2	22.5	25.1	26.4	26.1	22.7	16.3	11.7	10.8
	2%	WB	5.3	5.6	8.0	11.5	15.8	18.3	20.2	20.3	17.0	13.3	9.8	7.7
		MCDB	6.2	7.1	11.9	16.6	20.4	23.6	25.0	23.8	20.2	15.1	10.6	9.0
	5%	WB	4.0	4.3	6.4	9.9	14.4	17.0	19.3	18.8	15.6	12.2	8.6	6.2
		MCDB	4.9	5.6	9.1	14.2	18.9	23.1	24.1	22.1	18.3	13.7	9.4	7.2
	10%	WB	2.9	3.1	5.2	8.6	13.0	15.7	18.3	17.8	14.5	11.1	7.5	4.8
		MCDB	3.9	4.3	7.3	12.4	17.5	20.6	22.8	20.9	16.8	12.3	8.1	5.7
Mean Daily Temperature Range	5% DB	MDBR	6.5	7.6	10.0	12.7	13.4	13.0	12.4	11.7	11.1	8.5	5.5	5.8
		MCDBR	5.9	8.0	13.6	17.9	18.5	18.0	17.1	15.2	14.3	10.2	6.6	5.9
		MCWBR	5.4	6.4	9.3	11.4	10.9	10.2	9.1	9.6	9.6	7.8	5.9	5.4
	5% WB	MCDBR	5.8	7.0	12.4	15.5	15.3	16.0	14.2	13.3	13.1	9.2	6.2	5.8
		MCWBR	5.4	5.8	8.8	10.3	9.8	9.6	8.3	9.1	9.3	7.4	5.6	5.4
Clear-Sky Solar Irradiance	taub	0.286	0.320	0.338	0.376	0.373	0.362	0.389	0.381	0.349	0.340	0.313	0.274	
	taud	2.269	2.316	2.361	2.324	2.397	2.451	2.396	2.413	2.487	2.430	2.317	2.210	
	Ebn at Noon	569	715	812	833	862	877	841	821	797	689	528	457	
	Edh at Noon	47	71	88	108	107	103	107	98	78	64	46	37	
All-Sky Solar Radiation	RadAvg	0.36	0.96	2.46	4.03	5.26	5.75	5.20	4.17	2.83	1.28	0.47	0.25	
	RadStd	0.06	0.18	0.26	0.48	0.46	0.28	0.58	0.41	0.37	0.21	0.09	0.03	